42. Samples of two types of electric bulbs were tested for life (in hours) and the following data were obtained.

	TYPE-I	TYPE-II
Number of units	8	7
Mean of the samples (in hrs.)	1134	1024
Standard deviation of the samples (in hrs)	35	40

Test the hypothesis that the population means are equal at 5% level of significance.

43. The number of pages typed by 5 DTP – operators for 1 hour in the morning sessions are 10, 12, 13, 8, 9 and the number of pages typed by them in the afternoon are 11, 15, 12, 10, 8. Is there any significant difference in the mean number of pages typed?

44. An IQ test was conducted to 5 persons before and after they were trained. The results are given below:

Candidates	I	II	III	IV	V
IQ before training	110	120	123	132	125
IQ after training	120	118	125	136	121

Test whether any change in IQ at 1% level of significance.

45. The marks secured by 9 students in Statistics and that of 12 students in Business Mathematics are given below:

Marks in Statistics	65	74	64	58	60	67	71	69	75			
Marks in Business Mathematics	52	45	59	47	53	64	58	62	54	61	57	48

Test whether the mean marks obtained by the students in Statistics and Business mathematics differ significantly at 1% level of significance.

46. A test was conducted with 6 students before and after the training programme. Their marks were recorded and tabulated as shown below. Test whether the training was helpful in improving their scores.

Before training	100	160	113	122	120	105
After training	120	155	120	128	115	100

47. An experiment was conducted 144 times with tossing of four coins and the number of heads appeared at each throw are recorded.

No. of heads	0	1	2	3	4
frequency	10	34	56	36	8

Fit binomial distribution to the above data.

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